Power_Bi Internship Document

Task 1: Exploring the Link Between Pricing and Discounts

Objective:

The core goal of this task was to dig into the Snapdeal dataset and see if there's a meaningful connection between how a product is priced and the discount it carries. In e-commerce, you often wonder: do luxury items get deeper discounts to entice buyers, or are big sales reserved for budget-friendly impulse buys? By treating both price and discount as continuous variables, I aimed to uncover the underlying promotional logic used on the platform.

Dataset:

In line with the internship guidelines, I stayed consistent with the original Snapdeal training dataset. I focused specifically on three key metrics:

- **Current Selling Price (INR)**
- **Discount Percentage (%)**
- **Top Section** (used for categorical context)

Preparation Process:

Data is only as good as its cleanliness. Before jumping into the visuals, I ran the dataset through the Power Query Editor in Power BI to scrub any blank or invalid entries. I also ensured the data types were "clean"—setting the price to a proper INR currency format and verifying that discounts were correctly recognized as numerical percentages. This groundwork was essential to ensure the final scatter plot was both accurate and readable.

Visualizing the Data:

I chose a **Scatter Plot** for this analysis because it's the gold standard for spotting correlations. Unlike a bar chart that aggregates data, a scatter plot lets every single product "speak" for itself.

To make the chart functional, I mapped the **Selling Price to the X-axis** and the **Discount Percentage to the Y-axis**. I also integrated a **Category Slicer** (covering Accessories, Footwear, Clothing, etc.), allowing a user to zoom in on specific niches. This turns a static chart into a dynamic tool where you can see how strategies shift from, say, Men's Fashion to Kids' Wear.

Key Observations:

The results were actually quite interesting. Instead of a straight line showing a clear trend, the data was scattered:

- **No "One-Size-Fits-All" Rule:** There isn't a strong linear relationship. A high price tag doesn't automatically mean a massive discount.
- **Inconsistency:** We see deep discounts on both high-end and budget items.
- **Category Influence:** It became clear that discounting is likely driven more by the *type* of product than the price point itself.

Business Takeaways:

From a management perspective, this tells us that Snapdeal's sellers aren't just using a "price-tier" formula. Instead, they're likely reacting to inventory levels, seasonal trends, or competitor pricing within specific categories. For an e-commerce manager, the takeaway is clear: don't apply uniform discounts. Strategy should be tailored at the category level to be truly effective.

Conclusion:

Ultimately, Task 1 allowed me to move beyond simple data entry and into the realm of behavioral analysis. By using Power BI to visualize these complex relationships, I've created a framework that helps turn raw numbers into actionable business intelligence.

Task 2: Deep Dive into Product-Level Performance

Objective

The goal of Task 2 was to move beyond broad category trends and provide a **detailed, product-level diagnostic.** By creating a structured view of every item's price, discount, and rating, I aimed to give stakeholders the ability to inspect exactly how individual products are performing. This "micro-level" visibility is essential for identifying top-performers and spotting products that might need a pricing or promotional adjustment.

Dataset & Preparation

I utilized the **Snapdeal Products dataset**, specifically focusing on five core attributes: **Product Name, Price, Discount, Rating (detail),** and **Subcategory.**

To ensure the data was "analysis-ready," I performed several quality checks:

- **Numeric Integrity:** Verified that Price, Discount, and Rating were all formatted as decimal numbers.
- **Currency Formatting:** Set the Price column to reflect the correct currency (₹) for professional reporting.
- **Categorical Logic:** Developed a custom **Discount Band** to group individual percentages into meaningful business segments.

Technical Logic: Segmenting the Strategy

To make the data easier to navigate, I wrote a DAX SWITCH function. This takes raw, messy discount percentages and organizes them into four clear "Intensity Tiers." This allows a manager to filter for "Very High" discounts instantly, rather than searching through thousands of individual rows.

Calculated Column: Discount Band

```
Discount Band =
SWITCH(
   TRUE(),
   snapdeal_products[Discount] <= 10, "Low (0–10%)",
   snapdeal_products[Discount] <= 30, "Medium (11–30%)",
   snapdeal_products[Discount] <= 50, "High (31–50%)",
   "Very High (>50%)"
)
```

Visualization: The Power of Structure

For this task, I chose a **Table Visual**. While charts are great for spotting trends, a table is the gold standard when you need **absolute precision.**

Key Features of the Table:

- **Individual Row Inspection:** See the exact performance of every product name.
- **Totals Row:** Provides an immediate summary of the financial and performance footprint of the current selection.
- **Clarity:** Allows for direct, side-by-side comparison of Price vs. Customer Rating.

Interactivity: Putting the User in Control

To meet the internship requirements for a dynamic dashboard, I integrated two high-impact **Slicers**:

- **Discount Band Slicer:** Allows a user to instantly zoom in on products based on their promotional intensity (e.g., "Show me only items with >50% discounts").
- **Subcategory Slicer:** Enables a focused look at specific niches like "Footwear" or "Electronics" without the noise of the rest of the catalog.

When these slicers are used, the table updates in real-time, allowing for a "search and find" style of analysis that static reports simply can't offer.

Observations & Business Insights

The granular view provided some revealing insights into the Snapdeal ecosystem:

- **The Discount-Rating Independence:** Ratings don't always climb with discounts. Some "Low Discount" items have higher ratings than "Very High" discount items, suggesting quality matters more than a sale price.
- **Operational Visibility:** I could identify specific "Hero Products"—items that maintain high prices *and* high ratings—which are the most profitable for the business.
- **Pricing Gaps:** Significant price variations were discovered across subcategories, highlighting where the platform is most competitive.

Management Takeaway:

This table acts as a **Product Report Card.** It helps managers identify which items should be featured in marketing campaigns (high rating/high price) and which items might need to be cleared out (low rating/high discount).

Conclusion

Task 2 successfully delivers a professional, interactive product-level dashboard. By combining clean data preparation with a structured tabular layout, I have provided a tool that supports both day-to-day operational checks and long-term pricing strategy. This task completes the "micro" half of our analysis, ensuring no single product is lost in the averages.

Task 3: Value Analysis – Balancing Price vs. Customer Satisfaction

Objective

The core goal of Task 3 was to determine if **higher-priced product subcategories consistently receive better customer ratings.** In a perfect world, a premium price reflects premium quality, leading to higher satisfaction. By analyzing these two metrics side-by-side, I aimed to evaluate how pricing influences a customer's perception of "value" across the Snapdeal marketplace.

Dataset & Preparation

Using the Snapdeal Products dataset in Power BI, I focused on five key pillars: **Price, Rating (detail), Subcategory, Discount,** and a custom **Discount Band.**

To ensure the analysis was both accurate and readable, I performed the following preparation:

- **Data Cleaning:** Verified that Price and Rating were correctly formatted as decimal numbers to allow for mathematical averaging.
- **The "Top 15" Focus:** To prevent the chart from becoming cluttered and to focus on the most impactful data points, I applied a filter to show only the **Top 15 subcategories** based on Average Price.

Technical Foundation

To provide a fair comparison across subcategories with varying volumes of products, I developed custom DAX logic to power the dashboard.

Segmenting Strategy

I created a **Discount Band** to categorize products into meaningful business segments. This helps us see if high-priced items are "propped up" by deep discounts or if they stand on their own merit.

```
Discount Band =
SWITCH(
   TRUE(),
   snapdeal_products[Discount] <= 10, "Low (0–10%)",
   snapdeal_products[Discount] <= 30, "Medium (11–30%)",
   snapdeal_products[Discount] <= 50, "High (31–50%)",
   "Very High (>50%)"
)
```

Standardizing the Metrics

I used these measures to calculate the "center of gravity" for each subcategory:

- **Average Price:** AVERAGE(snapdeal_products[Price])
- **Average Rating:** AVERAGE(snapdeal_products[Rating (detail)])

Visualization Strategy: The Combo Chart

I implemented a **Combo Chart (Clustered Column + Line Chart)** to tell this story. This is the most effective way to compare two different scales—currency (INR) and ratings (1–5 stars)—simultaneously.

Configuration:

- **The Columns (Primary Y-Axis):** Represent the **Average Price**.
- **The Line (Secondary Y-Axis):** Tracks the **Average Rating**.
- **The X-Axis:** Displays the **Subcategory**.

Interactive Filtering:

To align with internship requirements for interactive dashboards, I added a **Discount Band Slicer.** This allows a user to filter the entire chart by discount intensity—for example, seeing if "High Discount" items have a different price-to-rating relationship than "Low Discount" items.

Key Observations & Business Insights

The data revealed a clear disconnect between cost and satisfaction:

- **Price $\neq$ Happiness:** Higher-priced subcategories do *not* automatically show higher ratings. In fact, many high-ticket segments showed only moderate satisfaction.
- **The "Value" Sweet Spot:** Several mid-priced categories showed remarkably strong customer ratings, suggesting they offer the best balance of quality and cost.
- **No Linear Correlation:** Because the rating line does not consistently follow the price bars upward, it is clear that price alone is not the driver of customer perception.

Strategic Takeaways:

From a business perspective, these findings suggest that **premium pricing must be backed by premium quality.** Simply increasing the price does not guarantee a better reputation. Instead, management should identify those mid-priced categories with high ratings and consider them for expansion, as they clearly resonate with the "value-seeking" customer.

Conclusion

Task 3 successfully demonstrates a comparative value analysis using a sophisticated dual-axis visualization. By integrating Top 15 filtering and a dynamic Discount Band slicer, I have created a tool that provides deep insights into how customers perceive value. This task strengthens the overall project by moving the conversation from "what we sell" to "how much value we provide."

Task 4: Relationship Between Discount and Product Rating

Objective:

In the final phase of this project, I set out to test a common e-commerce assumption: Do deeper discounts lead to better customer ratings? Intuitively, one might think a bargain makes a customer more forgiving, but in reality, the relationship is often more complex. By analyzing how discount levels impact satisfaction, I aimed to see if aggressive price-cutting is actually an effective tool for boosting a product's reputation.

Dataset & Logic:

Using the Snapdeal Products Dataset in Power BI, I focused on three primary pillars: **Discount, Rating and Subcategory.** To make the data more "human-readable," I moved beyond raw percentages and created a custom **Discount Band** using DAX. This allowed me to group products into logical buckets:

- **Low** (0–10%)
- **Medium** (11–30%)
- **High** (31–50%)
- **Very High** (>50%)

Technical Implementation (DAX):

To ensure the visualization was statistically accurate, I wrote a SWITCH function to categorize the discounts and created measures for **Average Discount** and **Average Rating**. This prevented individual outliers from skewing the overall category trends.

Visualization:

The Scatter Plot Strategy

I chose a **Scatter Plot** for this task because it excels at showing "clusters" of behavior.

- **The X-Axis** tracked the Average Discount.
- **The Y-Axis** tracked the Average Rating.
- **The Legend** was color-coded by my custom Discount Bands.

By plotting the data this way, each dot represents a subcategory. If the "more discount = more happy" theory were true, we would see a clear upward diagonal line. Instead, the dots were spread across the map, telling a much more nuanced story.

Key Observations:

The data effectively debunked the myth that big sales buy high ratings:

- **No Upward Trend:** There was no consistent "climb" in ratings as discounts increased.
- **The Mid-Range Winner:** Many subcategories in the "Medium" discount band actually outperformed those in the "Very High" band.
- **Diminishing Returns:** In several cases, products with massive discounts (>50%) had stagnant or even lower ratings, suggesting that customers might associate "too good to be true" prices with lower quality.

Business Interpretation:

For a marketing or category manager, these insights are crucial:

1. **Quality Over Price:** A discount can get a customer to *buy*, but only product quality will get them to *rate* it well.
2. **Protect the Brand:** Over-discounting can actually backfire by lowering the "perceived value" of the item. If it's always 70% off, customers start to wonder why.
3. **Strategic Promotion:** Discounts should be used to drive volume or clear inventory, but they shouldn't be relied upon as a tool to "fix" a low-rated product.

Conclusion:

Task 4 successfully proved that while discounts are a powerful lever for sales, they are not a shortcut to customer satisfaction. By visualizing this relationship through a scatter plot, I've provided a clear evidence-based argument for a more balanced promotional strategy. This final analysis rounds out the project by focusing on the most important metric of all: the long-term perception of the customer.

Task 5: Trend of Average Discount Over Time

Objective

In Task 5, I shifted from analyzing *what* is being sold to *when* the deals are happening. The goal was to uncover the "heartbeat" of Snapdeal's promotional strategy by tracking average discount levels across different scraping timestamps. By doing so, I aimed to identify whether these discounts remain consistent or if they fluctuate in response to time-based triggers, such as inventory updates or seasonal shifts.

Dataset & Preparation

I utilized the core Snapdeal Products Dataset, focusing specifically on the **Discount** values and the **Scraped At** timestamps.

Because time-series data is sensitive to formatting, I conducted a thorough preparation phase. I ensured the "Scraped At" column was recognized as a proper Time format in Power BI and validated that no blank entries would disrupt the flow of the trend line. This groundwork was essential to ensure the final visual accurately reflected the movement of the market.

The Analytical Foundation (DAX)

To add structure to this analysis, I utilized two key DAX components:

Segmenting Intensity

I brought back the **Discount Band** logic to categorize products into meaningful groups. This allows us to see if "Very High" discounts are a constant presence or if they only appear during specific windows.

```
Discount Band =
SWITCH(
   TRUE(),
   snapdeal_products[Discount] <= 10, "Low (0–10%)",
   snapdeal_products[Discount] <= 30, "Medium (11–30%)",
   snapdeal_products[Discount] <= 50, "High (31–50%)",
   "Very High (>50%)"
)
```

Performance Metric

I created an **Average Discount** measure to calculate the "mood" of the marketplace for any given moment on the timeline:

Average Discount = AVERAGE(snapdeal_products[Discount])

Visualization & Interactivity

I implemented a **Line Chart** to tell this story. The Line Chart is the undisputed standard for trend analysis because it excels at showing the fluid movement between data points.

Configuration:

- **X-Axis:** Scraped At (Time)
- **Y-Axis:** Average Discount
- **Continuous Axis:** Enabled to show a seamless progression of time.

To make the chart truly interactive and "internship-ready," I added two dynamic **Slicers**:

- **Subcategory Slicer:** To see how trends vary between, for example, Footwear and Electronics.
- **Discount Band Slicer:** To focus the trend line on specific levels of promotional intensity.

Key Observations & Business Insights

The visualization revealed that Snapdeal's pricing strategy is anything but static:

- **The Fluctuating Pulse:** Average discounts are not uniform; they show clear "peaks and dips." This suggests that the platform or its sellers are making active, time-based adjustments to their pricing.
- **Promotional Windows:** Specific intervals show significantly higher intensity, likely reflecting synchronized promotional updates.
- **Strategic Consistency:** By using the filters, we can see if certain categories maintain a steady discount rate while others are more volatile.

From a Management Perspective:

This temporal analysis is a powerful tool for **Dynamic Pricing.** It allows a manager to evaluate whether their promotional policies are being applied consistently and helps them identify the best "windows" for launching new campaigns based on historical patterns.

Conclusion

Task 5 successfully adds a layer of "Time Intelligence" to the project. By moving from simple averages to trend-based analysis, I have provided a roadmap for understanding how discount strategies evolve. This task completes the analytical depth of the dashboard, proving that in e-commerce, **when you offer a deal is just as important as the deal itself.**

Task 6: The Executive KPI Dashboard

Objective

The final stage of this project was to move beyond individual charts and create a centralized **Executive KPI Dashboard**. In a fast-paced e-commerce environment, stakeholders often need a "pulse check" on the business in seconds. This dashboard summarizes pricing, promotional intensity, and customer satisfaction into three clear metrics, while providing the interactive tools to slice that data by category or discount level.

Dataset & Preparation

Using the Snapdeal Products Dataset in Power BI Desktop, I focused on the four pillars of product performance: **Price, Discount, Rating,** and **Subcategory**.

To ensure the dashboard was both accurate and easy to navigate, I performed a "Data Health Check"—verifying currency formats for INR (₹), setting discounts to percentages, and ensuring ratings were numerically consistent.

Technical Engine (DAX & Logic)

To make the dashboard truly dynamic, I developed custom calculations. These aren't just static numbers; they are "filter-aware" formulas that update instantly based on user selection.

Categorizing the Strategy

I created a **Discount Band** to move away from raw percentages and into meaningful business segments. This allows management to see exactly how "Aggressive" vs. "Conservative" sales are performing.

```
Discount Band =
SWITCH(
   TRUE(),
   snapdeal_products[Discount] <= 10, "Low (0–10%)",
   snapdeal_products[Discount] <= 30, "Medium (11–30%)",
   snapdeal_products[Discount] <= 50, "High (31–50%)",
   "Very High (>50%)"
)
```

Powering the KPIs

I used the following measures to aggregate the data across the entire catalog:

- **Average Price:** Average Price = AVERAGE(snapdeal_products[Price])
- **Average Discount:** Average Discount = AVERAGE(snapdeal_products[Discount])
- **Average Rating:** Average Rating = AVERAGE(snapdeal_products[Rating])

Visualization & Interactivity

I implemented these metrics using **KPI Cards** for maximum visibility. Unlike complex graphs, cards are designed to shout the most important information at a glance.

The Current Snapshot:

- **Average Price:** ₹412.58
- **Average Discount:** 69%
- **Average Rating:** 3.13

Interactive Slicers: To meet the internship's requirements for a professional user experience, I added two primary interactive tools:

- **Subcategory Slicer:** Allows a user to zoom in on specific niches (e.g., "Footwear" vs. "Accessories").
- **Discount Band Slicer:** Allows a manager to filter by the intensity of the sale.

When a user interacts with these slicers, every KPI card on the page updates in real-time, transforming a static report into a conversational analytical tool.

Business Interpretation

From a strategic standpoint, these high-level figures reveal a specific story about the Snapdeal ecosystem:

1. **Market Positioning:** An average price of **₹412.58** suggests the platform is focused on the high-volume, mid-range consumer.
2. **Promotional Aggression:** A **69% average discount** is very high. While this drives sales volume, it puts heavy pressure on margins and brand perception.
3. **The Satisfaction Gap:** An average rating of **3.13** is a clear signal for quality control. It suggests that while the prices are attractive, the product experience isn't yet meeting "premium" expectations.

Conclusion

Task 6 successfully delivers a professional "Command Center" for product performance. By combining high-level KPIs with granular, interactive filtering, I've created a bridge between raw data and executive decision-making. This dashboard proves that data isn't just about recording what happened—it's about providing the tools to decide what happens next.